

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I _____V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1.5	1	10
Q2	2	3	2	1.5	1	9.5
Q3	2	3	2	1.5	1	9.5
Q4	2	3	2	1.5	1	9.5

Faculty Signature

Experiments

QUESTIONS:4

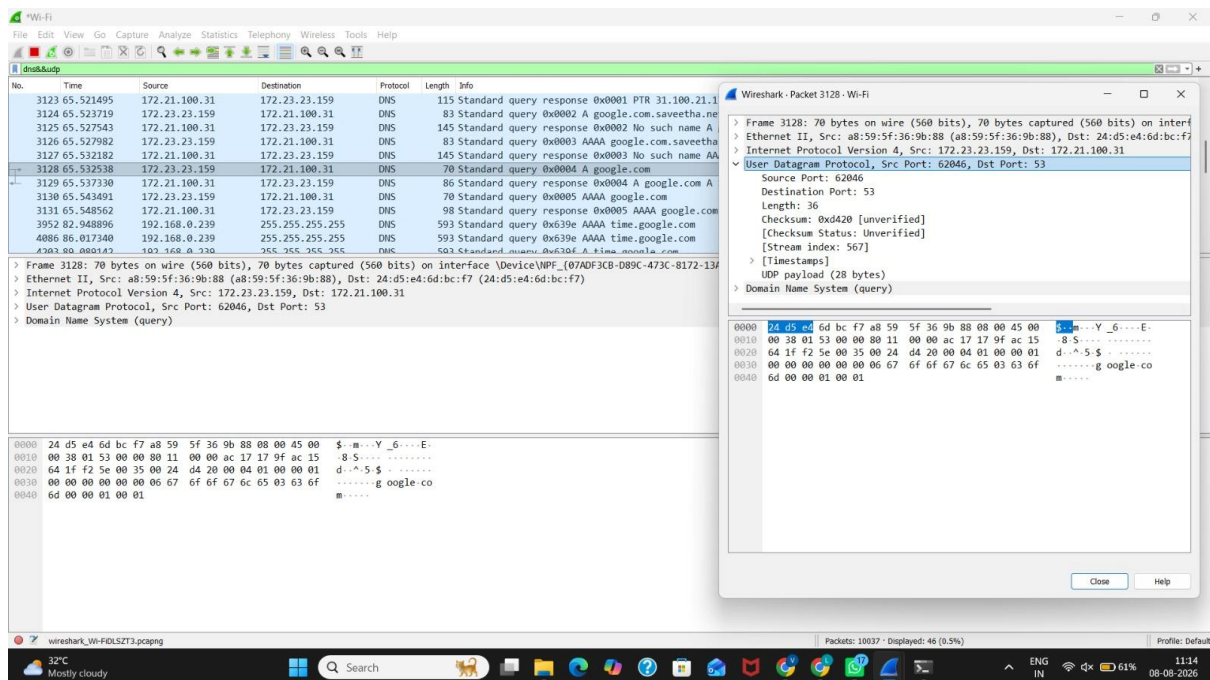
Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4)

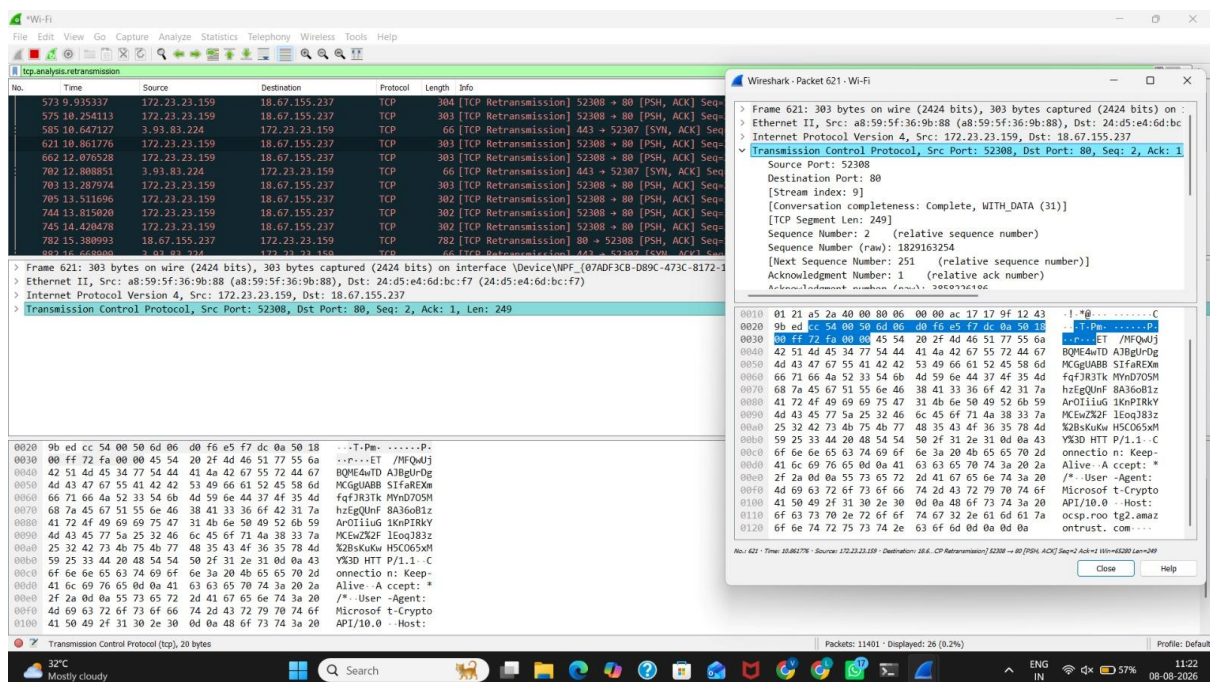
The screenshot shows a Wireshark capture of a TCP connection setup. The packet list on the left shows three packets: a SYN packet (Seq=555899), a SYN-ACK packet (Seq=555899, Ack=555899), and an ACK packet (Seq=555899, Ack=555899). The packet details pane on the right shows the details of the selected packet (Frame 137: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0). The packet is an Internet Protocol Version 4, Src: 172.23.23.159, Dst: 172.23.23.159. The Transmission Control Protocol details show: Source Port: 443, Destination Port: 64720, [Stream index: 2], [Conversation completeness: Incomplete (12)], [TCP Segment Len: 0], Sequence Number: 1 (relative sequence number), Sequence Number (raw): 4049026670, [Next Sequence Number: 1 (relative sequence number)], Acknowledgment Number: 60 (relative ack number), Acknowledgment number (raw): 954550892, 0101 = Header Length: 20 bytes (5), > Flags: 0x010 (ACK), Window: 903, [Calculated window size: 903], [Window size scaling factor: -1 (unknown)], Checksum: 0xc43 [unverified], [Checksum Status: Unverified], Urgent Pointer: 0, > [Timestamps], [Time since first frame in this TCP stream: 0.008307000 seconds], [Time since previous frame in this TCP stream: 0.008307000 seconds], > [SEQ/ACK analysis].

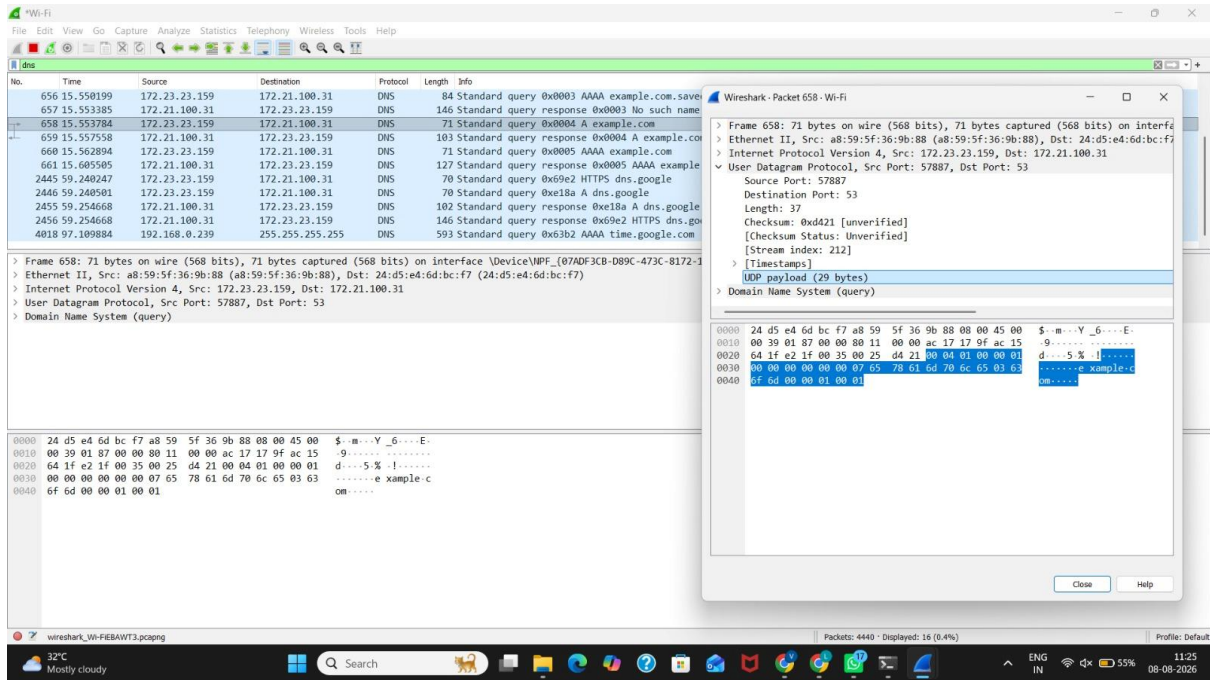
The screenshot shows a Wireshark capture of a TCP connection setup. The packet list on the left shows three packets: a SYN packet (Seq=61159), a SYN-ACK packet (Seq=61159, Ack=61159), and an ACK packet (Seq=61159, Ack=61159). The packet details pane on the right shows the details of the selected packet (Frame 352: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0). The packet is an Internet Protocol Version 4, Src: 172.23.23.159, Dst: 172.23.23.159. The Transmission Control Protocol details show: Source Port: 61159, Destination Port: 443, [Stream index: 5], [Conversation completeness: Incomplete, DATA (15)], [TCP Segment Len: 0], Sequence Number: 0 (relative sequence number), Sequence Number (raw): 51296253, [Next Sequence Number: 1 (relative sequence number)], Acknowledgment Number: 0, Acknowledgment number (raw): 0, > Flags: 0x002 (SYN), 0000 = Reserved: Not set, ...0 = Nonce: Not set, ...0 = Congestion Window Reduced (CWR): Not set, ...0 = ECH-Echo: Not set, ...0 = Urgent: Not set, ...0 = Acknowledgment: Not set, ...0 = Push: Not set, ...0 = Reset: Not set, > ...1 = Syn: Set, ...0 = Fin: Not set, [TCP Flags:S.], 0000 24 d5 e4 6d bc f7 a8 59 5f 36 9b 88 00 00 45 00 \$-m-Y_6---E- 0010 00 34 8c fd 40 00 80 06 00 00 ac 17 1f 9f 68 12 4-@---h- 0020 20 2f ee e7 01 bb 1e 98 47 7d 00 00 00 00 00 02 /-----G)----- 0030 ff ff 4c 1e 00 00 02 04 05 b4 01 03 03 08 01 01 -L----- 0040 04 02

2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4)

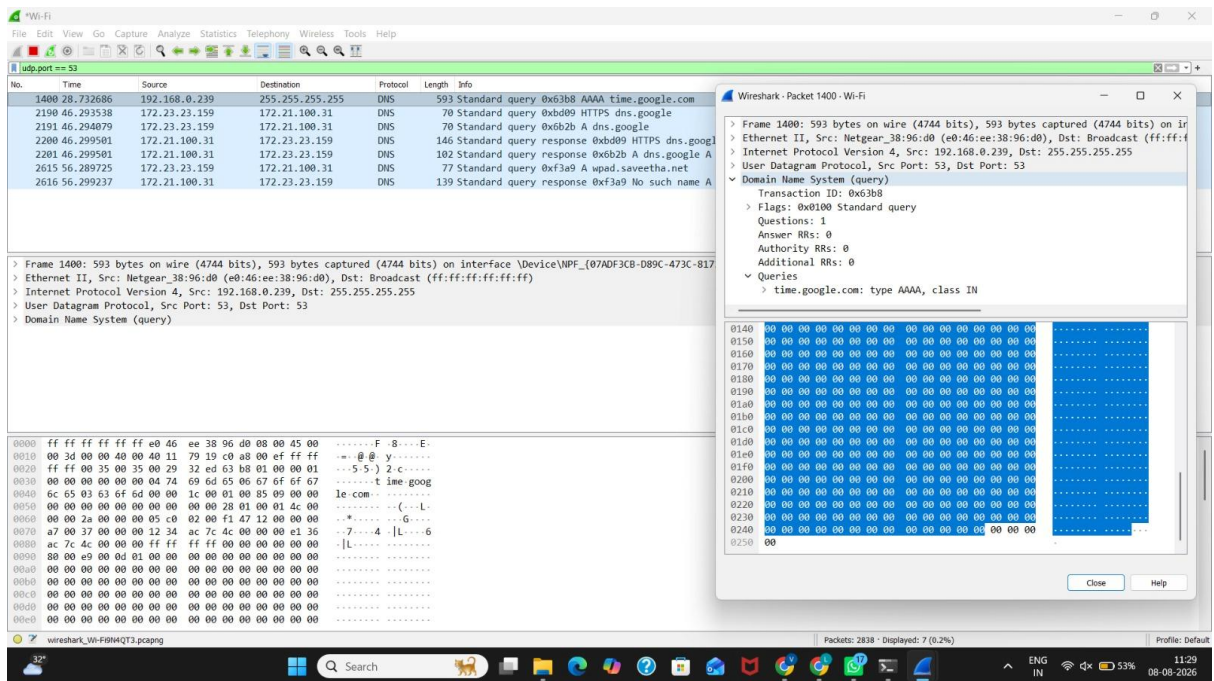


3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)





4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation